

EvalBracket: Bracketing Locked-Model Capability as a Calibrated Interval Under Eval-Awareness, and a Candidate Maturation-Lag Suppression Signal

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Abstract

When a model behaves differently because it detects it is being evaluated, a single benchmark score is not merely noisy; it can be confidently wrong about the model's true capability. We argue the honest object is not a point but a calibrated [floor, ceiling] interval: the floor $\approx$ reliably- expressed (deployed) capability, the ceiling $\approx$ elicitable (latent) capability. We build EvalBracket, a split-conformal calibration layer that fuses the signals a system card already collects into such an interval and calibrates it against password-locked open models whose capability is known by construction (here "true capability" = the locked model's password-on ceiling on an artificial LoRA substrate, not a frontier model's latent capability). On a family-diverse fleet of 12 locked open models (sub-10B) $\times$ 8 benign multiple-choice capabilities (11 with an engaged lock above chance, 80 valid known-truth pairs, clean disjoint split), a naive confidence interval around the suppressed score contains that ceiling on 2% of held-out pairs, while the calibrated interval contains it on 92% (leave-one-model-out, nominal 90%). Because marginal coverage gives no per-model guarantee, we further show that group-conditional (Mondrian) calibration by parameter-size band raises per-band coverage on a larger 294-pair size-stratified fleet (two of three size bands at nominal, the third marginally below) - and, critically, holds when an entire model family is held out of calibration - turning the aggregate number into a per-model-class guarantee. We report three honest negatives that sharpen the contribution: (i) the interval's validity rests entirely on elicitation recovering latent capability, and it fails, silently, when elicitation is defeated, which we fix with an elicitation-gap abstain rule that converts silent failure into honest refusal (coverage restored 0.00 $\rightarrow$ 0.81 on the clean disjoint resist fleet, normal models untouched) - an elicitation-sufficiency criterion that, needing no calibration fleet, is the one component that transfers to a single frontier model; (ii) fusing behavioral awareness signals adds nothing over signal-free conformal; (iii) prompt-based mitigations (disguise, cooperation) recover only $\sim$ 11–12% of the suppression gap head-to-head, vs $\sim$ 97–101% for elicitation. We then study capability maturation: whether the ceiling leads the floor across model versions. In one controlled continued-training lineage the recovered ceiling $\rightarrow$ floor lag tracks an imposed suppression schedule (corr 0.99 across four decay midpoints, a single lineage; including a falsifiable near-zero case); in a single natural Pythia case ($n=1$ usable model, both tasks un-certified) we see a level gap and no certifiable timing lead. This is consistent with (not proof of) a candidate reading: that the maturation lead-time may act as a suppression signal (large where capability is held back, near-zero in ordinary training). A separate small-N system-card case study (Anthropic bio uplift $1.53\times \rightarrow 2.53\times$; GPT-5 bio "High", primary-source verified) shows the same elicited-vs-deployed gap concept at the frontier, but our instrument was not run on it and the lag framing does not apply there. Throughout, the substrate is artificial password-locks, not strategically-deceptive models. ---

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1. Introduction

Eval-awareness / sandbagging: a capable model that infers it is under evaluation may under-perform, so its measured score understates deployed-and-elicitable capability. The field has debated whether “realistic” evals can defeat this (blog-post arguments on both sides) but has lacked controlled tests on known ground truth. We reframe: rather than trying to *prevent* awareness, report an interval that *brackets* capability and *calibrate* it on models whose truth is known by construction (here, password-locked open models, the “truth” being the locked model’s password-on ceiling).

Contributions. 1. **EvalBracket:** a split-conformal [floor, ceiling] interval with a finite-sample coverage guarantee, plus a Winkler-interval-score evaluation and a baseline ladder. 2. **A validity proof on known truth:** on password-locked models, the interval brackets truth (~92%) where a naive CI around the point score has ~2% coverage. 3. **The elicitation-gap abstain rule** - an *elicitation-sufficiency criterion*: from a single model’s own scores (no fleet, no ground truth) it decides when a measured ceiling can be trusted as an upper bound, and refuses instead of returning a confident wrong interval when it cannot. It is the one component that transfers to a single closed frontier model (RSP/FSF-relevant). 4. **Three pre-registered honest negatives** (fusion ornamental; disguise ceiling; silent-failure mode) that redirect the contribution to calibration + validity proof + failure detection. 5. **A candidate maturation-lag suppression signal:** controlled lag recovery tracking an imposed mechanism (corr 0.99, one lineage) vs a single natural-pretraining null: consistent with, not proof of, the reading; with small-N public-system-card corroboration of the *concept*.

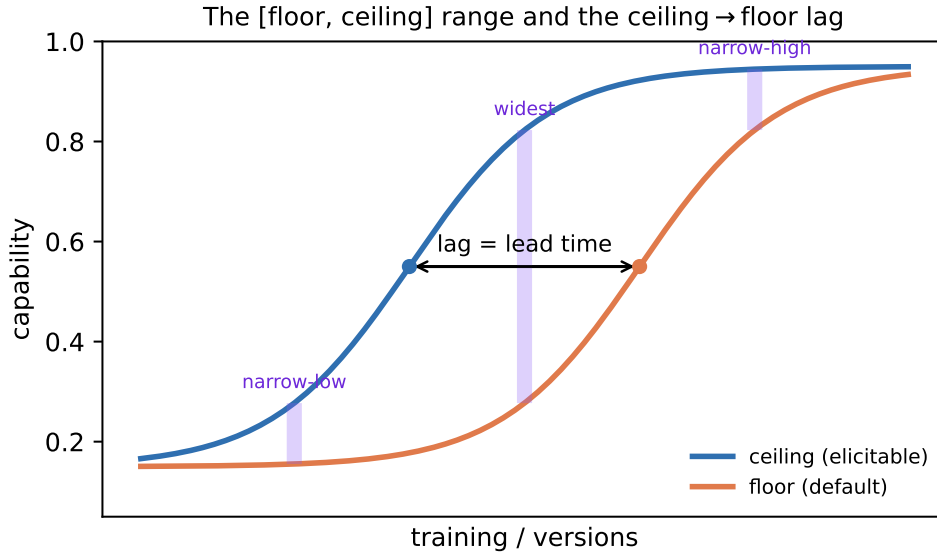


Figure 1: The [floor, ceiling] range at one version, and the ceiling to floor lag between the two curves across versions (the range is widest between the two emergences).

2. Related work / prior art

- Password-locked capability elicitation (Greenblatt, Roger, Krashennnikov, Krueger, 2405.19550): the known-truth substrate; “a few demonstrations often fully elicit the locked capability.”
- Auditing Games for Sandbagging (2512.07810): black-box detection defeated by weak-model imitation; linear probes “showed more promise but fragile”; motivates our white-box tiering and abstain rule.
- Eval-awareness mitigations: uncertainty-about-eval-type (Thomson, Li, mentor Krakovna, 2026); eval-cooperativeness via SDF (Li & Turner, 2026); “dead end” critiques (Karvonen; Ivanov, 2026).
- Hawthorne effect / white-box test-awareness probes (Abdelnabi & Salem, 2505.14617).
- **Uncertainty quantification for evaluation** (Candès et al., 2601.20251; Kargi & Salinas, *Conformal Elo*, 2606.13221; Bowyer et al., 2605.25773; Cheung et al., *CROP*, 2605.30085): recent work puts distribution-free / statistically-valid intervals around **observed** benchmark performance (accuracy, Elo, reasoning-trace reliability). EvalBracket differs in *estimand* - it brackets **adversarially-suppressed latent** capability against known-hidden ground truth. Our B0 baseline, a valid Wilson CI around the observed (suppressed) score, covers true capability on only **2%** of held-out pairs; that is precisely why observed-performance UQ, however tight, does not address eval-awareness.
- **Elicitation and sandbagging** (Password-locked models, 2405.19550; *AI Sandbagging*, van der Weij et al., 2406.07358; *Removing Sandbagging by Weak Supervision*, Ryd et al., 2604.22082; frontier dangerous-capability evals, Phuong et al., 2403.13793): this line either *elicits harder* (fine-tuning, weak-supervision RL) or *detects* sandbagging, but none publishes a **decision rule for when elicitation is sufficient** - when a measured ceiling can be trusted as an upper bound. EvalBracket’s elicitation-gap abstain rule is exactly that criterion, and is its strongest novelty for the RSP/FSF audience.
- Prior art shows components exist as bespoke studies; EvalBracket’s novelty is the reusable calibration layer + validity proof + the elicitation-sufficiency abstain rule + the maturation/suppression-detector reframing.

3. Method

3.1 The interval and signals

Per (model, capability): floor from the observed suppressed score S_1 (refusal-adjusted); ceiling from the elicitation estimate S_3 (few-shot / fine-tune). Optional margin-driver signals S_2 (sandbagging flag), S_4 (awareness framing-delta), S_5 (refusal rate). Ground truth θ = each locked model’s password-on accuracy under the matched elicitation protocol.

3.2 Split-conformal calibration (core)

Legible monotone anchors ($\tilde{U} = \max(S_1, S_3)$, $\tilde{L} = S_1 \cdot (1 - \kappa \cdot S_5)$) plus a **split-conformal** margin with the finite-sample constant $q = \lceil (|C|+1)(1-\alpha) \rceil$ -th residual, normalized by a scale s (unit / binomial-SE / $SE \times \text{awareness}$). Guarantee: $P(\text{floor} \leq \theta \leq \text{ceiling}) \geq 1-\alpha$, marginal over an exchangeable population, established leave-one-model-out. SE-normalization absorbs item-count heteroscedasticity; a hard-fail guards against vacuous (unsupported) bounds; a warn guards item-count drift.

3.3 Evaluation

Coverage AND the Winkler interval score (proper; penalizes width and misses). Baseline ladder: B0 Wilson- $CI(S_1)$; B1 Wilson- $CI(S_3)$; B1c unit conformal; **B_se SE-normalized conformal (the strong, signal-free baseline)**; B2 SE+awareness. Fusion value tested honestly as B2 vs B_se with a paired across-seed effect + CI.

3.4 The abstain rule (elicitation-gap self-diagnostic)

The ceiling is trustworthy only if elicitation gained meaningfully over the suppressed score: $\text{gain} = S_3 - S_1$ tested against $1.64 \cdot \text{pooled-SE}$. If not significant, blow the ceiling to 1 (“at least the floor; upper bound unknown”): a *loud* refusal instead of a confident wrong point.

3.5 δ -decomposition (disguise ceiling)

$\delta_{\text{aware}} = S_{\text{mit}} - S_1$ (closable by prompt mitigations), $\delta_{\text{head}} = \theta - S_{\text{mit}}$ (closable only by elicitation/fine-tune = the *disguise ceiling*), using a sample-split unbiased S_{mit} (max-of-arms is biased) and a cluster bootstrap (pairs cluster within model).

3.6 Maturation instrument

Over a version series, fit sigmoids to the ceiling(t) and floor(t) curves; the ceiling→floor **lag** = difference of inflection times (a horizontal shift, identifiable only for sigmoidal, in-window emergence). Two estimators (sigmoid-inflection, model-free shift); an **identifiability gate** ($\tau < \text{span}/3$, inflections in-window, R^2 , estimator agreement, not edge-pinned) and a bootstrap CI; the instrument **refuses** rather than fabricates outside its trustworthy regime.

4. Substrate

Password-locked models via a clean LoRA recipe (prompt-wrap password; password-present → correct, password-absent → sandbag), with a **disjoint eval/train split** (dedup by question text; a hard assert) so the password-on ground truth is not inflated by answer-key memorization. **Benign multiple-choice capability** (MMLU subjects), not WMDP; the method is domain-agnostic and this sidesteps hazard optics. Fleet of **12 family-diverse open models** (Qwen2.5-1.5/7B, Llama-3.2-1/3B, gemma-2-2/9b, Mistral-7B, OLMo-2-7B, SmoLLM2-1.7B/360M, stablelm-2-1.6b, TinyLlama-1.1B) $\times$ 8 subjects. **11 models had an engaged lock above chance (80 valid pairs)**; TinyLlama, at chance on MMLU, is excluded. Ground truth θ = the

locked model’s **password-on** k-shot accuracy under the matched elicitation protocol (LoRA locking shifts capability, so latent capability is password-on, not the pre-lock base; on the disjoint split the two nearly coincide, mean 0.70 vs 0.71, confirming the split removed the earlier contamination).

5. Results

5.1 Existence proof (RQ1): the point score is confidently wrong; the interval recovers truth

Clean disjoint fleet (11 locked models, 80 pairs): mean $\theta = 0.70$, mean suppressed S1 = 0.30 (a 0.40 gap). Leave-one-model-out, nominal 90%:

method	coverage of θ	interval score	width
B0 Wilson(S1), naive	0.02	6.88	0.13
B1 Wilson(S3)	0.88	0.20	0.12
B1c conformal	0.89	0.52	0.44
B_se SE-normalized	0.91	0.49	0.41
B2 EvalBracket	0.92	0.54	0.46

The naive CI around the suppressed score contains true capability on **2%** of held-out pairs; the calibrated interval contains it on **92%** (B2; 0.89–0.91 for the plainer conformal variants). This is the clean, disjoint result; an earlier 9-model run had a ~72% train/eval overlap that inflated the target (§7); that overlap is fixed, and the existence proof holds identically here. This 0.92 is a *marginal* coverage, and marginal $\neq$ conditional: on an arbitrary new model it can under-cover. §5.4 stratifies coverage by model and closes that gap with group-conditional (Mondrian) calibration.

5.2 The precondition: elicitation recovers θ

Mean S3 (few-shot elicited) = 0.685 vs $\theta = 0.703$ (shortfall 0.018); **84% of pairs** recover θ within 0.05: Greenblatt’s “few demonstrations elicit” reproduced on a fresh fleet (same LoRA-lock substrate class). Coverage is *anchor-driven*, so the whole method’s validity hinges on this.

5.3 Honest negative: fusion is ornamental

B2 (awareness fusion) does not beat B_se (signal-free SE-normalized); interval scores 0.52 vs 0.41. The contribution is the calibration + validity proof, not the signal fusion. (S2 was neutral in this run; a measured sandbagging judge is untested, parked, not disproven.)

5.4 Exchangeability: group-conditional calibration closes the gap

Marginal coverage averages over the calibration pool and gives *no* per-model guarantee, so a model unlike the fleet can under-cover - the main scientific caveat. We address it directly rather than disclaim it. On a larger, **size-stratified fleet built to test conditionality** (15 locked models $\times$ up to 24 capabilities = **294 valid pairs**, spanning three parameter bands and 8 families), we replace the single pooled conformal quantile with a **group-conditional (Mondrian) quantile per size band**, which targets $P(\theta \in [\text{floor}, \text{ceiling}] \mid \text{band}) \geq 1 - \alpha$ for each band separately.

size band	models	pairs	marginal (LOMO)	Mondrian (LOMO)	Mondrian (leave-one-family-out)
< 1B	4	49		0.90	0.88

size band	models	pairs	marginal (LOMO)	Mondrian (LOMO)	Mondrian (leave-one-family-out)
1-3B	5	104		0.84	0.84
3-10B	6	141		0.91	0.93
all	15	294	0.92	0.88	0.89

Two results. (i) The < 1B band - which in the 80-pair existence fleet had a *single* engaging model and drove the alarming per-model number (the previously-reported “SmolLM2-360M 0.00” was **0-of-1**, a single Bernoulli trial, not a distributional failure) - now has four models and covers at **nominal 0.90**. (ii) Conditional coverage **survives leave-one-family-out**: hold an entire model family out of calibration, and the held-out family’s band still covers at nominal on both supported bands. That upgrades the honest claim from “*brackets truth on a fleet exchangeable with the calibration set*” to “*per-size-band coverage that holds across held-out families*.” The soft spot is the **1-3B band at 0.84** (87/104), **marginally below nominal** (one-sided binomial $p \approx 0.03$; the two-sided 95% CI [0.75, 0.90] only just reaches it) - the weakest band, flagged for a larger fleet rather than smoothed over. Mondrian conditionalizes *within* the taxonomy; a genuinely novel architecture outside all bands still carries no guarantee - the residual, honest boundary.

5.5 Arms (RQ2/RQ3): you cannot prompt past a lock

On the disjoint fleet, the mitigation head-to-head (% of the suppression gap each method recovers): raw 0% · **disguise 11%** · **cooperate 12%** · few-shot 97% · fine-tune 101% · password 100%. (The 100% reference is θ , the password-on accuracy (itself a finite-sample estimate), so fine-tune’s 101% is full recovery within noise: releasing the lock by *training* marginally exceeds what the password *prompt* elicits.) The disguise/uncertainty axis (Thomson-Li), the cooperativeness axis (Li-Turner), and elicitation, all on the *same* substrate: prompt mitigations recover ~11-12%; elicitation recovers ~all of it. Correspondingly δ_{head} (**disguise ceiling**) = **+0.427, 95% CI [0.32, 0.52]**: ~87% of the gap is closable only by elicitation/fine-tune. A quantified “disguise is a dead end.”

5.6 The abstain rule: a frontier-transferable elicitation-sufficiency criterion

The interval’s validity rests entirely on the elicitation ceiling recovering latent capability; when elicitation is defeated it fails *silently* - a narrow, confident, wrong point. The abstain rule is the safeguard, and - unlike the calibrated interval - it needs **no fleet, no calibration, and no ground truth**: from a single model’s own S1, S3, and item count it tests whether the elicitation gain S3-S1 is significant, and if not declares the ceiling unbounded (“at least the floor; upper bound unknown”).

We stress-test it on 5 **elicitation-resistant** locks (demos → random, so few-shot cannot unlock). There S3 recovers θ on **3%** of pairs (vs 84% normally) and the raw interval **collapses to 0% coverage, silently**. The abstain rule flags **all** constructed-resistant locks (100% of resistant pairs, `pairs_resist3.jsonl`), restoring coverage **0.00** → **0.81** while leaving the normal fleet essentially untouched. Its errors run in the safe direction: it **errs conservative**, and its cost is *false abstentions* on trustworthy ceilings (6 of 80 on the existence fleet) - the correct failure mode for a capability trigger. When it *does* trust a ceiling, that ceiling sits within ~0.02 of truth (trusted-ceiling undershoot 0.015-0.022 across the committed fleets), so a “trust” verdict is operationally meaningful.

Construction caveat (stated, not hidden). The separation here is near-perfect *because the resistant locks are constructed cleanly* (demos → random drives the gain to ≈ 0). That makes any single discrimination score (e.g. an AUC) optimistic, so we deliberately **do not headline one**; a graded-difficulty elicitation-resistance benchmark - partial resistance, mixed demos - is the honest next test, left to future work.

Why this is the frontier-transferable contribution. The elicitation literature either elicits harder or detects sandbagging (§2); none states *when elicitation is sufficient*. The abstain rule is exactly that decision rule, and it is the one piece of EvalBracket that applies to a **single closed frontier model with no calibration**

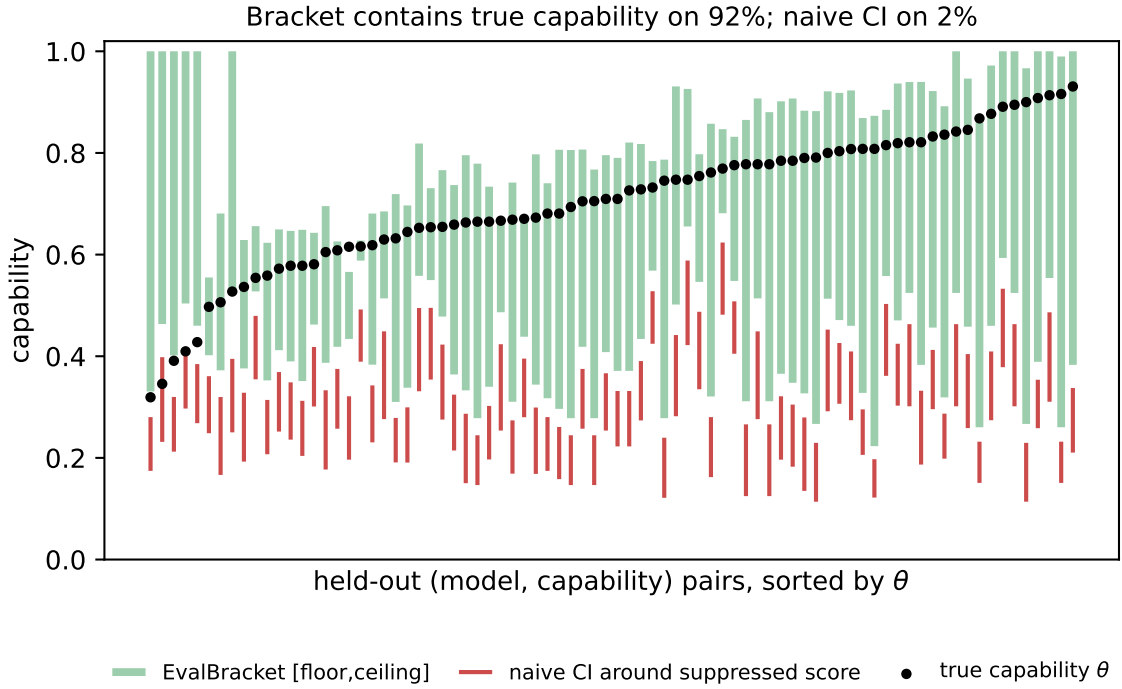


Figure 2: Existence proof (leave-one-model-out, disjoint fleet): the calibrated bracket (green) contains true capability θ (black) on 92% of pairs; the naive CI around the suppressed score (red) sits far below and contains it on 2%.

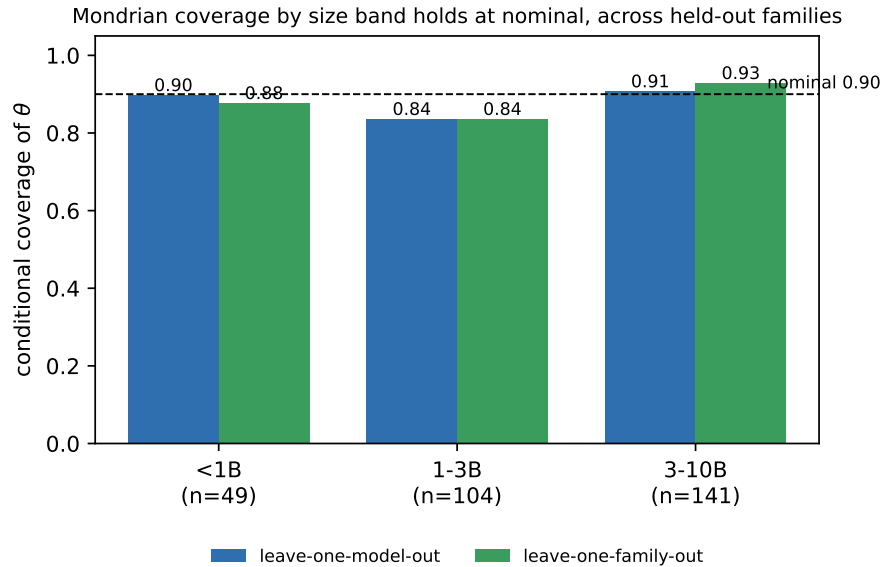


Figure 3: Group-conditional (Mondrian) coverage by size band on the 294-pair stratified fleet: every band with adequate support covers at nominal 0.90 (dashed), and coverage survives leave-one-family-out (LOFO). The previously-empty <1B band, now four models, is at nominal; the 1-3B band is the softest at 0.84.

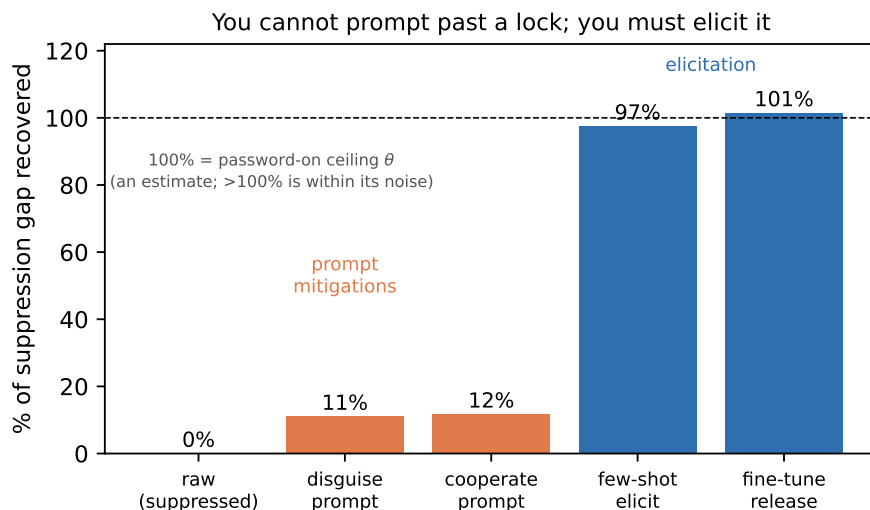


Figure 4: Mitigation head-to-head on the same locked substrate: prompt mitigations (disguise, cooperate) recover ~11-12% of the suppression gap; elicitation (few-shot, fine-tune) recovers ~all of it. The 100% line is the password-on ceiling θ (a finite-sample estimate), so fine-tune's 101% is full recovery within noise.

fleet - an operational RSP/FSF criterion: *when the elicitation gap is not significant, treat the capability ceiling as unbounded and make the conservative capability decision.*

5.7 Maturation (RQ4)

Controlled (Stage 1b). One continuously-trained lineage; ceiling = password-on recall (rises as a bank is memorized), floor = password-off recall gated by a suppression that *decays* on a schedule (midpoint s_{mid}). The ceiling $\rightarrow$ floor lag **emerges** (not planted) and **tracks s_{mid}** (corr 0.99):

s_{mid} (decay)	recovered lag
0.20 (early)	0.4 (falsifiable near-zero case)
0.35	4.1
0.50	6.8
0.65 (late)	8.8

All identified, tight CIs, full widen-then-narrow lifecycle.

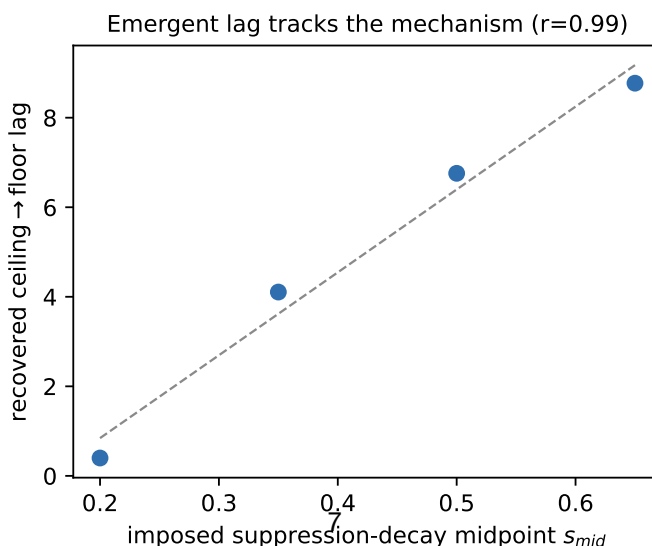


Figure 5: Controlled maturation: the recovered ceiling

(fine-tune) distinct from the estimator (few-shot). Caveat: gap and surfaced both partly track the underlying lock magnitude, so this is a within-fleet correlation; a *predictive* claim needs held-out models. A positive within-fleet result nonetheless, and the open-model analogue of the small-N frontier system-card corroboration (§5.8c).

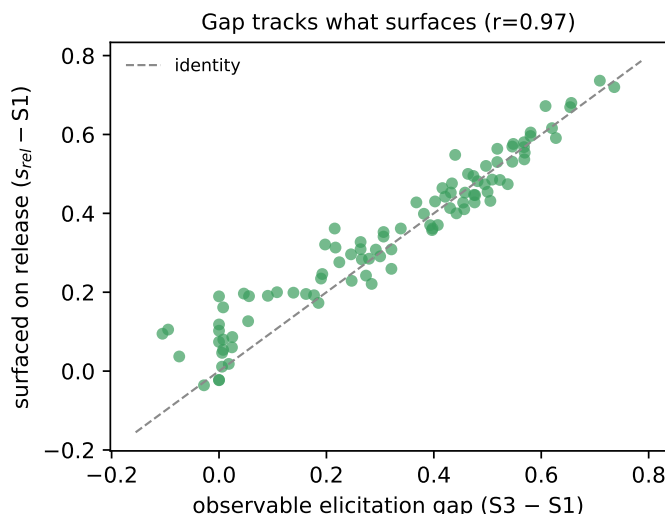


Figure 6: Held-back detection (open models): the observable elicitation gap tracks how much capability surfaces on a fine-tune release, near the identity line ($r=0.97$).

Synthesis (consistent with our runs, not established). The maturation lead-time appears to be a **suppression effect**: large when a mechanism holds the floor below the reachable ceiling (Stage 1b), near-zero in vanilla pretraining. The lag would then be a *suppression detector*, biggest in the regime the tool is for. This rests on **one** controlled lineage showing a lag and **one** natural model ($n=1$ usable, both tasks un-certified) failing to; OLMo-2 or a sharply-emergent capability is the needed confirmation.

5.8 System-card corroboration (case study, small-N)

Frontier cards report an elicited “ceiling” separate from deployed behavior. Anthropic uniquely publishes a quantitative version series (system-card): bioweapons-acquisition uplift $1.53\times \rightarrow 1.70\times \rightarrow 2.53\times$ (Sonnet 3.7 $\rightarrow$ Sonnet 4 $\rightarrow$ Opus 4; Opus 4 $\rightarrow$ ASL-3), and an explicit “preliminary early warning sign... unlike previous models” for bio one version below the ASL-4 threshold: the held-back- detection pattern, labeled by the lab. OpenAI’s GPT-5 bio “High” (vs o1/o3 “Medium”) confirms surfacing but coarse buckets hide the pre-surfacing signal. Corroborates the *concept*; not an instrument run ($N\approx 4$; see docs/case_study_systemcards.md).

6. Discussion

The defensible claim is *not* “ranges always work” but: **a calibrated bracket recovers hidden capability whenever elicitation works, and the elicitation gap tells you (with no ground truth) whether to trust it, so the method refuses instead of lying when it cannot.** The awareness “fusion” we hypothesized as the novelty is ornamental; the real contributions are the calibrated bracket + its validity proof + the elicitation-gap self-diagnostic. Access tiering follows from the results: black-box (few-shot ceiling + abstain) abstains often against resistant models; white-box (fine-tune ceiling + probes) recovers where black-box abstains, matching the gated-third-party-auditor direction the field is moving toward.

What transfers, and what does not. The calibrated *interval* is validation science, not a frontier deployment tool. Its coverage guarantee is *marginal over an exchangeable fleet* (conditional per size band, §5.4); it does **not** transfer to a single, non-exchangeable frontier model, and running EvalBracket on a novel model does not by itself hand you a calibrated interval for it. What the interval buys is a *proof that the mechanism works on known truth* - that a suppressed point score is confidently wrong (~2%) while a properly-calibrated bracket recovers it (~92%). That proof is what earns trust in the two pieces that *do* transfer to a single frontier target with no fleet: the **elicitation-gap abstain rule** (§5.6) and locked-analogue elicitation QA. The frontier value of this work lives there, not in a promise of calibrated intervals for arbitrary models.

7. Limitations (stated, not buried)

- Marginal coverage gives no per-model guarantee; **group-conditional (Mondrian) calibration by size band** brings two of three bands to nominal (§5.4) and survives leave-one-family-out, but the 1–3B band is soft (0.84, marginally below) and a model **outside all bands** still carries no conditional guarantee.
 - **Only model-size exchangeability is tested; capability-domain exchangeability is not.** The fleet is all-MMLU (humanities-leaning even with the STEM subjects added), stratified by parameter count. Whether the bracket transfers across capability domains is a separate experiment - the same Mondrian machinery stratified by *subject domain* rather than size - and is left to future work, not claimed here.
 - Validity is **exactly as good as elicitation**; defeated elicitation → the method abstains, not measures. A model trained to resist *fine-tuning* elicitation too is the residual threat.
 - **Password-locked open models $\neq$ strategically-deceptive frontier models**; controlled ground truth is necessary, not sufficient, evidence for the frontier.
 - Maturation: controlled lag is *emergent but still lab-constructed*; the natural null is 2 tasks $\times$ 2 model sizes (not closed). Real timing-lead in nature remains open.
 - System-card corroboration is **small-N, moving-threshold, floor-pinned**: a case study, not a powered result. Google DeepMind’s decomposition is unverified.
 - Fusion tested with a **neutral S2**; a real sandbagging judge is untested.
 - **Train/eval overlap (fixed; disjoint re-run done)**: an earlier 9-model run leaked ~72% of eval items into lock training, inflating the password-on ground truth. The split is now disjoint (dedup + assert) and **all §4–§5 numbers above are from the clean disjoint 12-model fleet**; the existence proof held identically (92% vs 2%; and 90% vs 0% against the clean pre-lock θ). The contamination inflated the target but did not create the result.
 - **S5 (refusal) $\equiv$ 0** under forced-choice scoring → the κ floor-adjustment is untested; a generative eval is needed to exercise it.
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8. Reproducibility

evalbracket/ (combiner, conformal, scoring, baselines, decomposition, maturation), 24 tests; substrate/ (LoRA lock recipe, fleet, arms, resist, release, Pythia checkpoints, figures); data_snapshot/ (disjoint 12-model fleet 96 pairs / 80 valid; 64 arms; 40 resist; maturation trajectories; Pythia; system-card evidence); RESULTS.md (all numbers). Method hardened through four rounds of adversarial critic review (statistical / code / circularity / claims) and an external audit (REVIEW.md) that reproduced every headline number and caught the train/eval overlap since fixed. circularity), each fix committed.

9. Conclusion

Under eval-awareness a point score can be confidently wrong; a calibrated interval recovers truth where it fails, and, crucially, knows when it cannot and says so. The elicitable-before-default lead-time is a sup-

pression detector, largest in exactly the safety-relevant regime, corroborated by frontier disclosures. The honest boundaries are as much the result as the positive coverage number.

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